

Daniel Kuo

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EDUCATION

Rice University

Houston, TX

Bachelor of Science in Computer Science, GPA: 3.79

Aug. 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Compilers, Computer Networks, Parallel Computing, Distributed Program Construction, Large-Scale Data Analytics

EXPERIENCE

Goldman Sachs

Dallas, TX

Software Engineering Intern, Asset & Wealth Management

June 2026 – Aug. 2026

- Re-architected the **Java** authentication layer behind a client-onboarding flow serving **35+ client organizations**, moving a statically stored credential out of version-controlled configuration into a **Sybase**-backed store with scheduled automatic refresh; decoupled rotation from code deployment, cutting recovery for a recurring **~8-hour production incident** to a **sub-10-second** database write.
- Designed rotation around **overlapping validity** rather than cutover, refreshing at the **6-month mark** of a **12-month** credential lifetime while retaining the prior credential as a live fallback, so in-flight requests authenticate against either credential and rotation incurs **zero downtime** by construction.

Labshare

Houston, TX

Co-founder & COO (Primary Contributor, Full-Stack Engineering / PM)

Jan. 2025 – Nov. 2025

- Created the product roadmap from interviews with **15+** lab managers to pilot launch with **3 labs**; served as primary contributor to the full-stack architecture in **Go** and **React Native**, defining the service layer, API contracts, and **MongoDB** schema design while running sprint planning, task breakdowns, and daily standups.
- Built a serverless **AWS Lambda/Step Functions** ETL pipeline using **LlamaIndex** and **Pinecone** to normalize inventory data into **MongoDB**; implemented **Atlas Search** with compound indexes for typo-tolerant lookup, cutting inventory overhead by **80%** for **30 research labs**.
- Deployed a **Go** server and **MongoDB Atlas** database across **4 AWS regions** with **CloudFront** caching and **Nginx** load balancing over an EC2 cluster for high availability; shipped the inventory search UX in **TypeScript** with **Zustand** and **TanStack Query**, featuring debounced queries and optimistic updates.

PROS

Houston, TX

Software Engineering Intern, Group Sales Optimizer (GSO) Team

May 2025 – Aug. 2025

- Designed and implemented a historical quote retrieval and replay feature for the **Group Sales Optimizer (GSO)** platform, building a **Java** backend with **Hibernate** and **Oracle** to surface previously internal-only data to clients, reducing a multi-hour log investigation to a three-click workflow.
- Led team to first place in the PROS Ideathon, pitching a **Jira Assistant** that used **Azure OpenAI** for action-plan generation, **vector databases** for semantic ticket search, and the **Jira API** to automate workflows and reduce MTTR.

PROJECTS

Project Plant | Go, Kafka, PostgreSQL, Redis, Next.js, Docker, k6

Apr. 2026 – June 2026

- Architected a distributed IoT telemetry platform (**Go** ingestion API, **Kafka** event buffer, Go stream processor, **Redis/Postgres** CQRS) sustaining **500+ events/sec** at **<50ms p99 latency**, verified via automated **k6** load tests gating CI.
- Built a hardware-agnostic Device Adapter SDK and a real-time **Next.js**/WebSocket fleet dashboard (Redis Pub/Sub fan-out, **<100ms p99** reads under write load); drove **TDD** across Go, Python, and TypeScript with testcontainers integration tests and full CI/CD dockerization.

2.5D Parallel Matrix Multiplication | C++, MPI, OpenMP

Apr. 2026

- Implemented a hybrid **MPI + OpenMP** 2.5D matrix multiplication library in **C++**, combining Cannon's algorithm with SIMD-vectorized, cache-blocked GEMM to reach **503 GFLOPS** across 72 ranks.
- Diagnosed a socket memory-bandwidth bottleneck via profiling and rebalanced MPI rank/OpenMP thread layout, raising strong-scaling efficiency from **53% to 72%** and cutting runtime by **1.7x** at scale.
- Overlapped communication with computation using persistent non-blocking MPI shifts over a custom Cartesian topology, verified correct to machine epsilon across all valid process-grid configurations.

DermAtlas | FastAPI, React Native, PostgreSQL, Vertex AI, GCP

Jan. 2026 – Apr. 2026

- Bootstrapped CI/CD (**GitHub Actions**) and a **TDD** test suite for DermAtlas, a HIPAA-aligned dermatology clinical-decision-support tool built by a 7-person team; logged the team's highest commit/file count (**29 commits, 155 files**) across frontend, auth, and CI.
- Built a synchronized pan/zoom lesion-comparison view in **React Native Reanimated** (shared gesture state across two images); led a security audit that caught **PHI** leaking into URL parameters and refactored client navigation to fix the HIPAA violation before launch.

TECHNICAL SKILLS

Languages: Java, Python, Go, JavaScript/TypeScript, C/C++, Haskell

Frameworks & Libraries: React/React Native, Node.js, Next.js, FastAPI, Unity, TanStack Query

Cloud & Developer Tools: AWS, Microsoft Azure, PostgreSQL, MongoDB, Sybase, Redis, Kafka, Docker, Git, NGINX, Postman, Cloudflare, Linux

AI Tools: Claude Code, Cursor, GitHub Copilot, Google Gemini, Google Antigravity, Ollama, LangChain, LlamaIndex, Vertex AI, Azure OpenAI, Pinecone, RAG